A Zero-Inflation Impact Economy

A Rule-Based Framework for Long-Run Price-Level Stability and Contribution-Linked Distribution

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Abstract

This paper proposes a Zero-Inflation Impact Economy (ZIIE), a monetary and distributive framework in which the long-run target for the general price level is zero inflation while individual relative prices remain free to adjust to scarcity, productivity, technology, and preferences. The framework does not impose literal price freezes. Instead, it combines (i) a price-level stabilization rule, (ii) state-contingent monetary adjustment based on real output and money velocity, and (iii) an optional contribution-linked distribution mechanism that allocates a bounded share of newly created money according to measurable time, effort, skill scarcity, and verified impact. The proposal is motivated by the cumulative erosion of purchasing power under permanently positive inflation targets and by theoretical literature showing that optimal inflation in several monetary models can lie near zero or below zero. It explicitly confronts the main objections to zero inflation: the effective lower bound on nominal interest rates, downward nominal wage rigidity, measurement bias, debt-deflation dynamics, supply shocks, and the difficulty of measuring social impact. The paper develops equations, governance constraints, transition rules, falsifiable hypotheses, and an agent-based simulation protocol comparing 0%, 0.01%, 1%, and 2% inflation targets. ZIIE is presented as a testable institutional hypothesis, not as a claim that zero inflation is universally optimal.

1. Introduction

Most major advanced-economy central banks currently define price stability around a positive inflation target. The U.S. Federal Reserve reaffirmed a 2% longer-run PCE inflation objective in its 2025 strategy statement, while the European Central Bank likewise maintains a symmetric 2% medium-term target. Their stated rationales include anchoring expectations, preserving room for interest-rate cuts, reducing deflation risk, accommodating downward nominal rigidities, and allowing for measurement bias.

A positive target, however, also implies deliberate long-run growth in the nominal price level. At exactly 2% annual inflation, a fixed unit of currency has only about 67% of its initial purchasing power after 20 years and about 55% after 30 years, assuming the target is continuously realized. This paper asks a narrower question than whether existing institutions are 'good' or 'bad': can a modern economy be designed around approximately constant long-run purchasing power without creating larger losses through unemployment, deflation, financial instability, or policy impotence?

The proposal separates two ideas that are often conflated. Zero inflation means stability of an aggregate price index over the relevant horizon; it does not mean that every product must always carry the same sticker price. Relative prices must remain capable of transmitting information about scarcity and productivity. The core target is therefore $\Delta P/P \approx 0$ for the general price level, not $\Delta P_i = 0$ for every good i.

The second component addresses distribution. Income in a market economy already reflects a mixture of time, bargaining power, ownership, scarcity, skill, risk, and measured output. ZIIE explores whether a limited, auditable component of monetary distribution can be linked more explicitly to verified contribution without replacing markets or allowing a central authority to assign arbitrary 'human worth' scores.

2. Related Monetary-Policy Debate

The case for positive inflation is substantial and must be treated as the benchmark rather than dismissed. The ECB explicitly identifies a safety margin against deflation and the effective lower bound, smoother macroeconomic adjustment, downward wage rigidity, and possible positive measurement bias as reasons for a 2% target. The Federal Reserve similarly argues that well-anchored 2% expectations support price stability, moderate long-term rates, and the ability to promote maximum employment during disturbances.

At the same time, the theoretical optimum is not settled at 2%. Schmitt-Grohé and Uribe survey leading monetary models and report optimal inflation rates ranging from the negative of the real interest rate to values insignificantly above zero. Other work finds mild deflation can be Ramsey-optimal under particular assumptions. Agarwal and Kimball argue that institutional changes that relax the zero lower bound could permit movement toward a zero inflation target. Conversely, IMF work incorporating endogenous growth and downward wage rigidity finds parameterizations in which the welfare-maximizing target exceeds 2%. The correct target is therefore an empirical and institutional question, not a theorem.

ZIIE should consequently be evaluated against competing regimes using identical shocks and welfare criteria. The burden of proof is on the zero-inflation design to demonstrate that purchasing-power stability is not purchased at the cost of persistent unemployment or financial fragility.

3. Definition of the ZIIE Regime

Let P_t denote a broad consumption price index, Y_t real output, M_t a monetary aggregate, and V_t money velocity. The primary long-run target is:

$$\pi^* = \Delta \ln(P_t) = 0$$

Operationally, exact zero is neither measurable nor controllable at every instant. The proposed regime therefore uses a narrow tolerance band:

$$-\varepsilon \leq \bar{\pi}_t,H \leq +\varepsilon$$

where $\bar{\pi}_t,H$ is average inflation over a rolling horizon H and ε is a small tolerance chosen after accounting for index measurement error. A candidate research calibration is ε = 0.05 percentage points, but this is a simulation parameter rather than a policy recommendation.

Individual prices remain flexible:

$$\Delta P_i,t \neq 0 \text{ is permitted while } \Delta P_t/P_t \approx 0$$

A drought can raise food prices; an improvement in semiconductor productivity can lower computing prices. ZIIE attempts to stabilize the weighted aggregate rather than suppress the signals contained in relative prices.

4. Monetary Stabilization Rule

The quantity identity provides a transparent starting point:

$$M_t V_t = P_t Y_t$$

Taking approximate growth rates gives:

$$\pi_t \approx gM,_t + gV,_t - gY,_t$$

Under a zero-inflation objective, the baseline money-growth rule is:

$$gM,_t^* = \hat{g}Y,_t - \hat{g}V,_t$$

Because output and velocity are estimated with error, policy should not mechanically implement this identity. A feedback version is:

$$gM,_t = \hat{g}Y,_t - \hat{g}V,_t - \varphi\pi(\hat{\pi}_t - \pi^*) - \varphi P \ln(P_t/P^*)$$

Here P^* is the target price-level path, which is flat in the strict ZIIE case; $\varphi\pi$ and φP govern the response to inflation and accumulated price-level drift. The price-level term is important: an inflation target can forgive past misses, whereas a price-level target requires overshoots and undershoots to be gradually corrected.

The rule can be implemented through several instruments—interest rates, reserve remuneration, open-market operations, standing facilities, fiscal-monetary transfers, or an electronic-money architecture. The paper deliberately distinguishes the target from any single instrument.

5. Contribution-Linked Income and Distribution

The distributive component begins with a decomposition of earned compensation rather than an attempt to calculate the intrinsic value of a person. For participant i:

$$Y_i = B_i + \alpha T_i + \beta E_i + \gamma I_i + \delta S_i + R_i$$

where B_i is a baseline or contractual component; T_i verified productive time; E_i an effort/difficulty proxy; I_i verified impact; S_i skill scarcity/responsibility; and R_i market, entrepreneurial, capital, or risk-based returns. The model does not require all income to be centrally determined.

If the monetary authority determines that $\Delta M_t > 0$ can be introduced consistently with the price-level target, only a policy-defined fraction θ of that increment is eligible for contribution-linked distribution:

$$F_t = \theta \cdot \max(\Delta M_t, 0), \quad 0 \le \theta \le 1$$

A raw contribution score can be constructed as:

$$C_i,t = wT\cdot T_i,t + wE\cdot E_i,t + wI\cdot I_i,t + wS\cdot S_i,t$$

and normalized:

$$s_i,t = C_i,t \,/\, \Sigma_j C_j,t$$

so that:

$$D_i,t = s_i,t \cdot F_t$$

This mechanism must be capped, contestable, privacy-preserving, and independently audited. It should reward verifiable contribution, not political loyalty, popularity, identity, belief, or obedience. A robust implementation should use multiple evidence channels and should make the scoring formula public.

6. Preventing the Impact Metric from Becoming a Social-Credit System

The impact mechanism is the most institutionally dangerous part of the proposal. Goodhart's law implies that once a measure becomes a target, agents will attempt to optimize the measure rather than the underlying social outcome. ZIIE therefore requires constitutional-style constraints:

- No single universal lifetime score. Scores are task- and period-specific and expire.
- No scoring of political, religious, ideological, or private lawful behavior.
- Individuals can inspect the evidence used, challenge errors, and appeal decisions.
- Multiple independent evaluators and statistical fraud detection replace a single ministry or company.
- The contribution-linked pool is bounded by θ; ordinary wages, entrepreneurship, contracts, ownership, and voluntary exchange continue to operate.
- Negative scores cannot confiscate previously earned lawful wealth merely because an evaluator dislikes an outcome.
- Impact is preferably measured through observable outputs and downstream outcomes rather than self-reported effort.
- Weights and algorithms are public, versioned, auditable, and subject to periodic democratic or constitutional review.

7. Supply Shocks and Relative-Price Adjustment

A zero aggregate inflation target cannot prevent real scarcity. Suppose an energy shock raises the efficient relative price of energy. Attempting to force the energy price itself back to its old nominal value would suppress information and risk shortages. Under ZIIE, energy is allowed to rise. Monetary policy responds only to the extent that the shock propagates into persistent generalized price growth.

The operational target should therefore distinguish first-round relative-price changes from second-round inflation persistence. A trimmed or median inflation measure can be used as a cross-check, while the official target remains a broad household price index. The system's success criterion is not 'nothing ever becomes more expensive'; it is that generalized monetary depreciation does not become the default long-run state.

8. The Hard Cases

8.1 Effective Lower Bound

At zero expected inflation, equilibrium nominal interest rates are lower than under a 2% target for a given real neutral rate, leaving less conventional room to cut rates during severe recessions. ZIIE therefore cannot be credible without an explicit lower-bound architecture. Candidate mechanisms include deeply negative reserve rates with an electronic-money standard, temporary fiscal transfers, state-contingent balance-sheet policy, or a temporary emergency price-level path. The simulation should test each mechanism rather than assuming the lower bound away.

8.2 Downward Nominal Wage Rigidity

Workers and firms may resist nominal wage cuts even when real adjustment is required. Positive inflation can facilitate real-wage reductions without explicit nominal cuts. A zero-inflation regime must therefore permit more flexible contracts, profit sharing, variable compensation, hours adjustment, retraining, and automatic sectoral reallocation. If unemployment rises persistently under realistic wage rigidity, the zero target fails its own welfare test.

8.3 Debt and Deflation

Unexpected deflation raises the real burden of nominal debt and can destabilize leveraged balance sheets. ZIIE targets zero inflation, not deflation, and uses price-level correction to prevent a persistent downward drift. Long-duration debt could additionally be indexed to the aggregate price level, reducing arbitrary transfers between debtors and creditors caused by deviations from the target.

8.4 Measurement Error

No consumer price index perfectly measures the cost of living. Quality change, substitution, new goods, housing, and sampling create uncertainty. The target band ε should therefore be no tighter than the statistical system can credibly measure. The proposal's 'zero' is an economic objective of stable purchasing power, not a claim of infinite measurement precision.

9. Welfare Objective

The regime should be judged by a transparent loss function rather than inflation alone:

$$L = \lambda\pi\pi^2 + \lambda P[\ln(P/P^*)]^2 + \lambda u(u-u^*)^2 + \lambda y(y-y^*)^2 + \lambda f F^2 + \lambda q Q^2$$

where u is unemployment, y the output gap, F a financial-stress measure, and Q a distributional distortion or inequality metric. A zero-inflation regime is accepted only if its reduction in inflation and price-level drift does not produce unacceptable increases in unemployment, lost output, instability, or distributional harm.

10. Agent-Based Simulation Design

The empirical contribution of the project should be a reproducible agent-based model rather than a purely verbal argument. A baseline economy can contain 10,000 households, heterogeneous firms, commercial banks, a fiscal authority, and a monetary authority.

Component	Baseline	Heterogeneity / Rule
Households	10,000	Skills, wealth, debt, consumption baskets, labor supply
Firms	500	Productivity, sector, markup, inventories, wage contracts
Banks	10	Capital, deposits, lending standards, defaults
Sectors	10-20	Food, energy, housing, manufacturing, services, technology, etc.
Monetary regimes	4	Targets: 0%, 0.01%, 1%, 2%
Horizon	30-50 simulated years	Monthly or quarterly time steps
Impact pool θ	0 to 0.25	Sensitivity analysis
Shocks	Common across regimes	Supply, demand, productivity, banking, energy, pandemic-like

Each regime receives the same random seed and shock sequence. Reported outcomes should include cumulative price-level drift, real GDP per capita, unemployment, consumption volatility, real wage dispersion, wealth concentration, debt-service stress, bankruptcies, financial crises, policy-rate lower-bound frequency, and welfare loss.

The central comparison is not whether 0% produces a prettier inflation chart. It is whether the entire welfare distribution improves after accounting for stabilization costs.

11. Experimental Hypotheses

- H1: A price-level-targeting zero-inflation regime preserves long-run purchasing power more effectively than a 2% inflation-targeting regime by construction.
- H2: Without an effective-lower-bound workaround, a zero target experiences larger recession losses after sufficiently large demand shocks.
- H3: With an effective lower-bound workaround and credible price-level targeting, the welfare gap between zero and positive inflation targets narrows materially.
- H4: Downward nominal wage rigidity determines a threshold below which reducing the inflation target raises unemployment costs.
- H5: Contribution-linked distribution can reduce concentration of monetary seigniorage benefits when impact measurement is accurate, but can reduce welfare when the metric is noisy or gameable.
- H6: A small operational band around zero outperforms literal instantaneous zero because it avoids excessive policy reaction to measurement noise.
- H7: Indexing long-term debt to the target price level reduces debt-deflation instability under a zero-inflation regime.

12. Transition from a Positive-Inflation Regime

An abrupt shift from 2% expected inflation to 0% could reprice bonds, wages, debt contracts, and asset valuations. ZIIE therefore requires a preannounced transition. One candidate path reduces the target by 0.25 percentage points per year until zero, conditional on inflation expectations, unemployment, and financial-stress thresholds. New long-duration public debt can be issued with price-level indexation during the transition.

Credibility is essential. The monetary authority must publish its target price-level path, estimation methods, policy rule, emergency escape clauses, and a procedure for returning to the path after shocks. An escape clause without a return rule would convert the framework into discretionary inflation targeting.

13. Institutional Architecture

A practical implementation separates powers. A statistical authority measures prices and output; a monetary authority stabilizes the price level; an independent audit network validates contribution metrics; courts or tribunals adjudicate disputes; and the fiscal authority determines taxes and ordinary public expenditure. No institution should simultaneously control the price index, create money, calculate individual impact, and adjudicate appeals.

The monetary authority's mandate can be summarized as: preserve the long-run purchasing power of the unit of account while supporting maximum sustainable employment and financial stability, subject to a published price-level target and emergency rules.

14. What Would Falsify the Proposal?

A serious proposal must specify failure conditions. ZIIE should be rejected or materially revised if, under realistic calibrations and repeated shock simulations, it generates persistent unemployment materially above comparison regimes; materially higher crisis frequency; recurrent debt-deflation spirals; extreme policy volatility required merely to hold the index near zero; or contribution scoring that remains highly gameable despite governance safeguards. Evidence that the welfare-maximizing inflation target is robustly positive under plausible parameters would also count against the strict zero target.

Conversely, evidence favoring ZIIE would require more than lower measured inflation. It would require comparable or better employment and output stabilization, lower cumulative purchasing-power erosion, acceptable financial stability, and a governance mechanism whose distributional benefits survive adversarial testing.

15. Discussion

The main conceptual claim of this paper is modest but important: permanent positive inflation should be treated as a policy choice with benefits and costs, not as the definition of price stability itself. A zero long-run target is theoretically defensible in some classes of monetary models, while other models give strong reasons for a positive buffer. The relevant research question is which institutional design performs best under realistic frictions.

The contribution-linked component is separable from the zero-inflation component. This modularity is useful. Simulations can first compare monetary targets while holding distribution constant; then activate the impact mechanism and estimate its incremental effects. If contribution scoring fails, the price-level framework can survive independently.

The framework also changes the political economy of monetary expansion. Rather than assuming that all new money must enter through one conventional channel, ZIIE makes the entry mechanism an explicit design variable. That opens a research program on whether monetary injections can be made more neutral, broad-based, or contribution-sensitive without compromising price stability.

16. Conclusion

This paper develops a testable blueprint for an economy targeting approximately zero long-run inflation while retaining flexible relative prices. Its central rule stabilizes a target price level; its monetary-growth benchmark responds to real output and velocity; and its optional contribution mechanism distributes only a bounded portion of permissible monetary expansion according to auditable measures of productive contribution.

The proposal does not assume that zero inflation is automatically superior to 2%. Instead, it converts that normative intuition into a falsifiable economic model. The next research stage is computational: calibrate a heterogeneous-agent economy, impose identical shocks under 0%, 0.01%, 1%, and 2% regimes, and identify the conditions under which stable purchasing power improves or worsens social welfare. A successful zero-inflation model must solve—not ignore—the lower bound, wage rigidity, debt-deflation, measurement, and governance problems.

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Appendix A. Minimal Computational Specification

At each time step t:

- Firms choose production, prices, vacancies, wages, and inventories using adaptive expectations and sector-specific productivity.
- Households choose labor supply and consumption subject to income, wealth, debt, and heterogeneous consumption baskets.
- Banks price credit and update capital after defaults.
- The statistical authority estimates P_t, Y_t, V_t, unemployment, and financial stress with realistic publication noise and revisions.
- The monetary authority applies the regime-specific rule.
- If $\Delta M_t > 0$, the model routes money through the configured injection mechanism; in ZIIE experiments, a fraction θ can enter the contribution pool.
- All agents update expectations. Identical exogenous shocks are applied across monetary regimes.

Appendix B. Illustrative Purchasing-Power Arithmetic

For constant annual inflation π, the purchasing power of one nominal currency unit after n years relative to today is:

$$PP(n) = 1 / (1 + \pi)^n$$

Inflation target	10 years	20 years	30 years
0.00%	100.0%	100.0%	100.0%
0.01%	99.9%	99.8%	99.7%
1.00%	90.5%	82.0%	74.2%
2.00%	82.0%	67.3%	55.2%